



ViM and Basic Unix command

NCSV102

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- **Definition:** Bash (Bourne Again Shell) is a Unix shell and command language.
- **Common uses:** File manipulation, program execution, and text processing.

Basic Bash Commands

- **pwd**: Print Working Directory
Example: **\$ pwd**
- **ls**: List directory contents
Example: **\$ ls -l**
- **cd**: Change Directory
Example: **\$ cd /home/user**
- **mkdir**: Make Directory
Example: **\$ mkdir new_folder**
- **rm**: Remove files or directories
Example: **\$ rm file.txt**
- **cp**: Copy files or directories
Example: **\$ cp source.txt destination.txt**
- **mv**: Move/rename files or directories
Example: **\$ mv oldname.txt newname.txt**

- **cd** to navigate directories:
Example: **\$ cd /home/user/documents**
- Use **ls** to list contents:
Example: **\$ ls -a**
- Display current directory with **pwd**:
Example: **\$ pwd**
- **Absolute vs. relative paths:**
 - Absolute path: **/home/user/documents**
 - Relative path: **/documents**

- Creating files with **touch**:
Example: **\$ touch newfile.txt**
- Viewing files with **cat**, **less**, **more**:
 - **cat newfile.txt**
 - **less newfile.txt**
 - **more newfile.txt**
- Introduction to vim for editing files

1. Basic Navigation and Directory Management:

- Create a new directory named **project** in your home directory.
- Navigate into the **project** directory.
- Create three subdirectories inside **project** named **src**, **bin**, and **docs**.
- Navigate back to your home directory using a relative path.

2. File Operations:

- Inside the **src** directory, create a new file named **main.py**.
- Copy **main.py** to the **bin** directory.
- Rename **main.py** in the **src** directory to **app.py**.
- Delete the **bin/main.py** file.

3. Listing and Viewing Files:

- List all files and directories in the **project** directory with detailed information.
- Use the **cat** command to view the contents of **src/app.py**.
- Use the **less** command to view the contents of a large file in the **docs** directory (create a large dummy file if needed).

4. Combining Commands:

- Create a new directory named **backup** in your home directory and copy all files from the **src** directory into **backup** in one command.

What is VIM?

- **Definition:** VIM (Vi IMproved) is a highly configurable text editor.
- History and evolution from Vi.
- **Key features:**
 - Syntax highlighting
 - Undo/redo
 - Search and replace
- **Why use VIM?**
 - Efficient text editing
 - Powerful features for developers

- Command to open a file: **vim filename**
- **Basic modes:**
 - Normal: Esc
 - Insert: i, a, o
 - Visual: v, V
 - Command: :

- **Normal Mode:** Navigation and commands
Navigation: h, j, k, l
- **Insert Mode:** Editing text
Enter Insert Mode: i, a, o
- **Visual Mode:** Selecting text
Enter Visual Mode: v, V
- **Command Mode:** Running commands
Enter Command Mode: :

- **Insert Mode:**
 - i: Insert before cursor
 - a: Insert after cursor
 - o: Open a new line below
- **Normal Mode:**
 - Navigation: h, j, k, l
 - Delete: x, dw, dd
 - Copy and paste: y, p
 - Undo and redo: u, Ctrl-R
- **Saving and quitting:**
 - Save: **:w**
 - Quit: **:q**
 - Save and quit: **:wq**
 - Quit without saving: **:q!**

- **Moving within a file:**

- gg: Go to the beginning
- G: Go to the end
- w: Move forward a word
- b: Move backward a word

- **Searching:**

- /search_term: Search forward
- ?search_term: Search backward
- n: Next occurrence
- N: Previous occurrence

- **Deleting text:**
 - x: Delete character
 - dw: Delete word
 - dd: Delete line
- **Copy and paste:**
 - y: Yank (copy)
 - p: Paste
- **Undo and redo:**
 - u: Undo
 - Ctrl-R: Redo

Yank (Copy) Text

- yy: Yank (copy) the current line
- 2yy: Yank (copy) two lines (change 2 to any number for more lines)
- yw: Yank (copy) the word under the cursor
- y\$: Yank (copy) to the end of the line from the cursor
- yG: Yank (copy) from the current line to the end of the file

Visual Mode:

- v to enter visual mode, then use arrow keys or hjkl to select text.
- Press y to yank (copy) the selected text.

Put (Paste) Text

- p: Put (paste) the yanked text after the cursor
- P: Put (paste) the yanked text before the cursor

1. Basic Text Editing:

- Open a new file named **notes.txt** with VIM.
- Enter Insert mode and type the following text:

This is a practice file.

We are learning VIM.

VIM is a powerful text editor.

- Save and quit the file.

3. Copy and Paste:

- Yank (copy) the second line and paste it after the third line.
- Yank the word VIM from the third line and paste it at the end of the fourth line.

4. Deleting Text:

- Delete the first line of the file.
- Delete the word powerful from the fourth line.

5. Search and Replace:

- Search for the word VIM in the file.
- Replace the first occurrence of VIM with Vim.
- Replace all occurrences of VIM with Vim.

6. Using Visual Mode:

- Enter Visual mode and select the first two lines.
- Copy the selected lines and paste them at the end of the file.
- Enter Visual Line mode and select the last two lines.
- Delete the selected lines.

4. Creating and Editing Files:

- Use Bash to create a file named **script.sh** in the **project** directory.
- Open **script.sh** with VIM.
- In VIM, enter the following Bash script:

```
#!/bin/bash  
echo "Hello, World!"
```
- Save and quit the file.
- Use Bash to make **script.sh** executable.
- Run **script.sh** using Bash.

- **GCC (GNU Compiler Collection):** The standard tool to compile C code on Linux.
- **Basic Compilation:**
 - Command: **gcc filename.c**
 - Result: Creates an executable file named **a.out** by default.
- **Compiling with a specific output name:**
 - Command: **gcc filename.c -o output_name**
 - Example: **gcc hello.c -o hello**
 - Result: Creates an executable named **hello**.
- **Check for errors:** If there are syntax errors, gcc will list them with line numbers.

- To run a program, you must use the **current directory** prefix: `./`
- **If you compiled without a name:**
Command: `./a.out`
- **If you named the output (e.g., hello):**
Command: `./hello`
- **Why `./` ?**
It tells the shell to look for the program in the *current directory*, not in the system folders.

- **Stopping a Program (Infinite Loop):**
 - If your program gets stuck or runs forever:
 - Press **Ctrl + C** to force-quit the running program.
- **Clearing the Screen:**
 - Command: **clear** (or Press **Ctrl + L**)
 - Useful when the terminal gets cluttered.
- **Using History:**
 - Use the **Up Arrow** key to see and reuse previous commands (avoids retyping).

Final Exercise: Your First C Program

1. Create & Edit:

- Create a directory **lab1** and go inside it.
- Use VIM to create a file named **welcome.c**.
- Type a basic C program:

```
#include <stdio.h>
int main() {
    printf("Welcome to C Lab at IIT ISM!\n");
    return 0;
}
```

- Save and quit (:wq).

2. Compile & Run:

- Compile: **gcc welcome.c -o run_me**
- Execute: **./run_me**

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Thank You for Listening!

